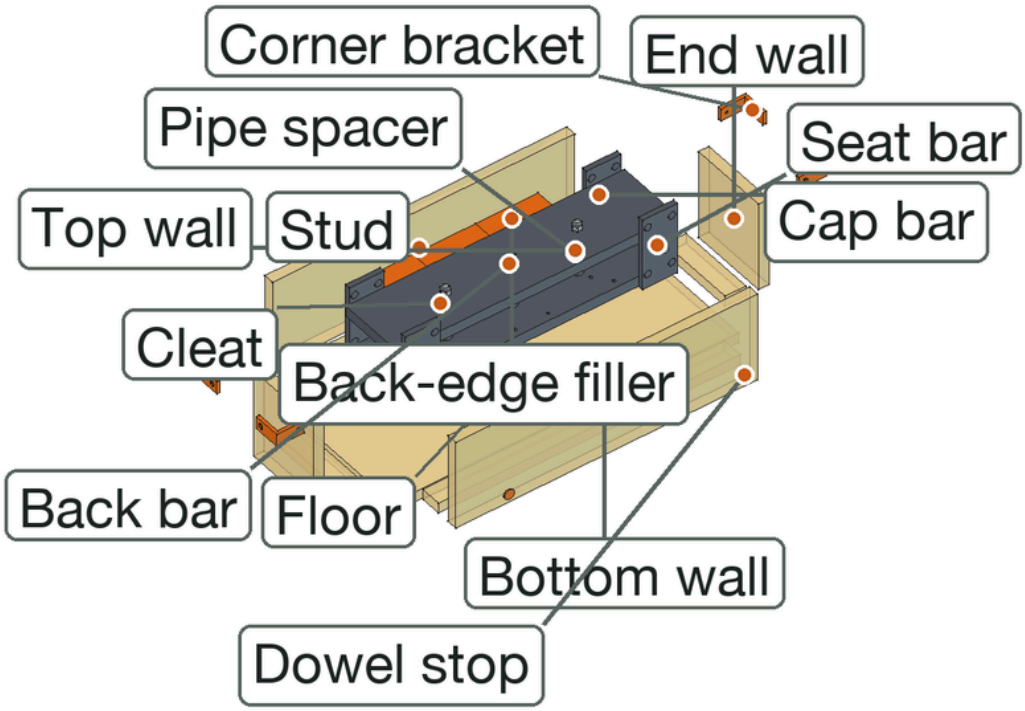
Beam mold

DIY CNC epoxy-granite mill · October 2026

How to make the beam mold. Drilling templates: `templates/beam-templates.pdf`. The sheet layout for the boards is `docs/src/boards/sheet-1.svg`; `docs/src/make_pdfs.py` also makes `molds-complete.pdf`, all the molds in one (not kept in the repo).

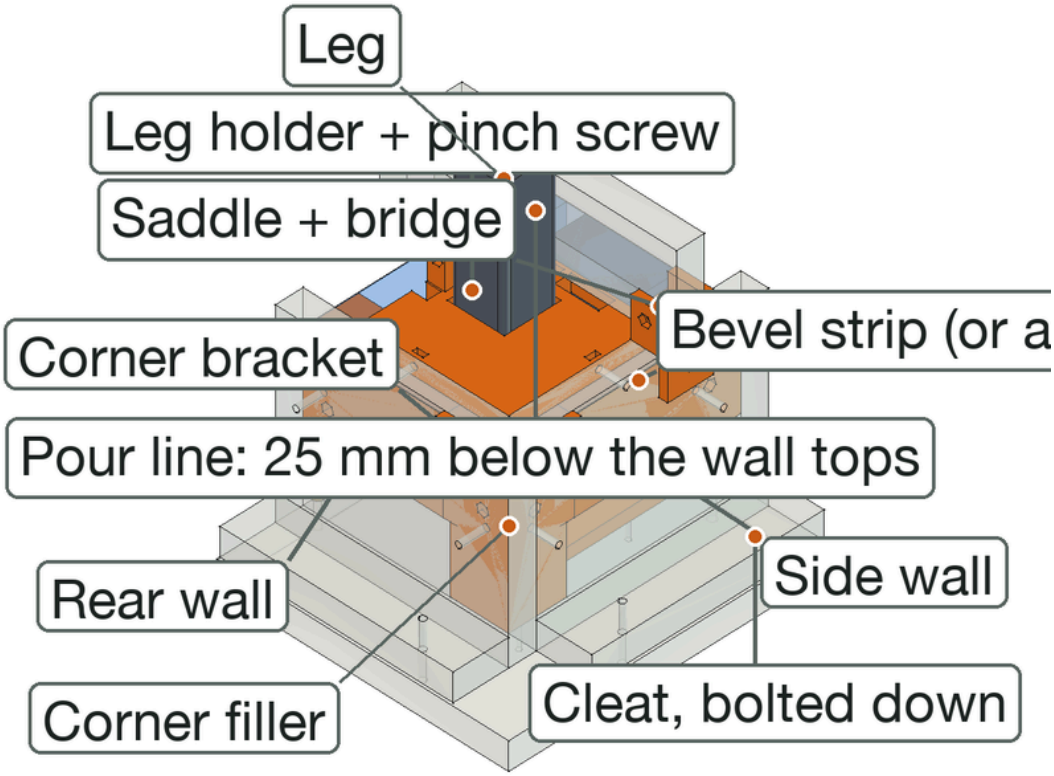


In short: melamine boards bolt together with printed brackets. Every printed piece bolts or pins to its board from outside. The walls stand 25 mm above the pour line, so no bolt goes through a face the mix touches. Nothing is glued, and it all comes apart for the next casting.

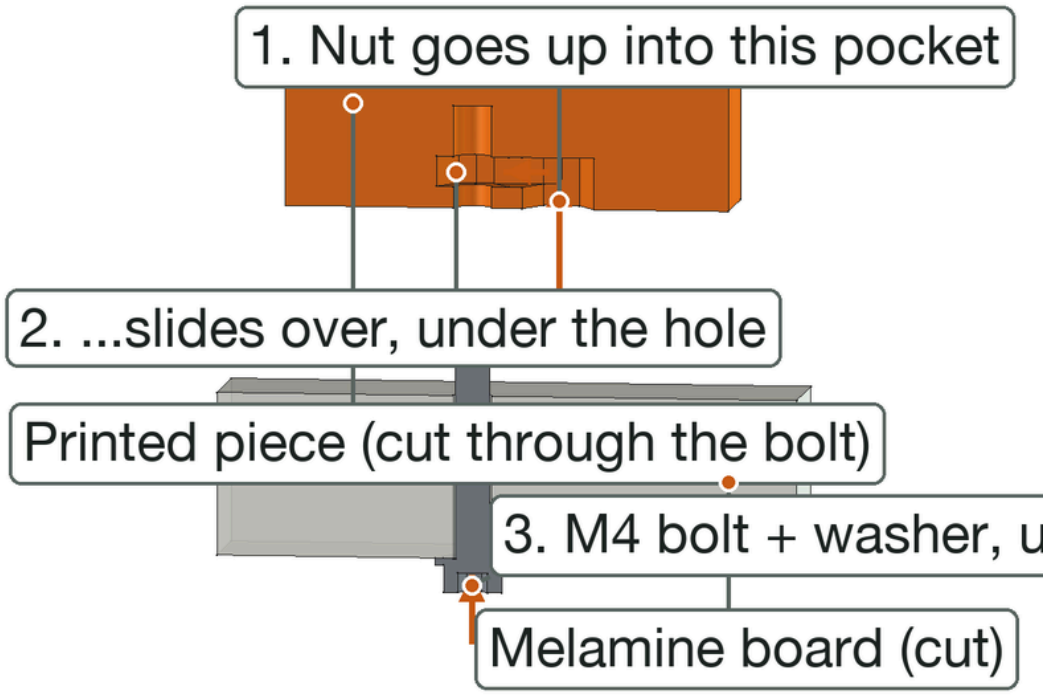
Order of work: print → cut the boards → mark them with the templates and drill → wax → assemble → pour → take apart.

1. How it works

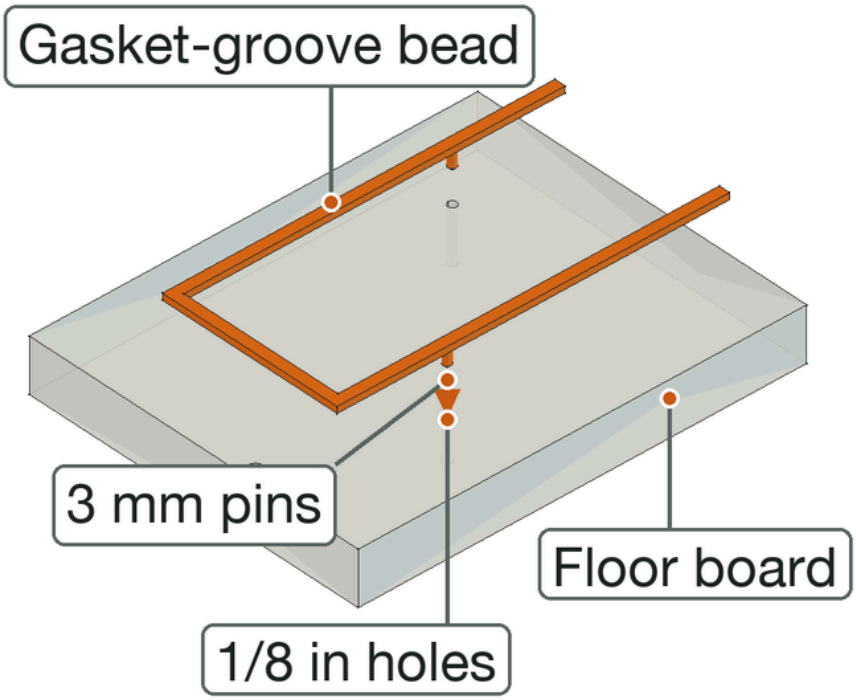
- **Walls** stand on the floor against bolted-down **cleats**; **corner brackets** square the tops. All the bolts go through the top 25 mm, above the pour line.



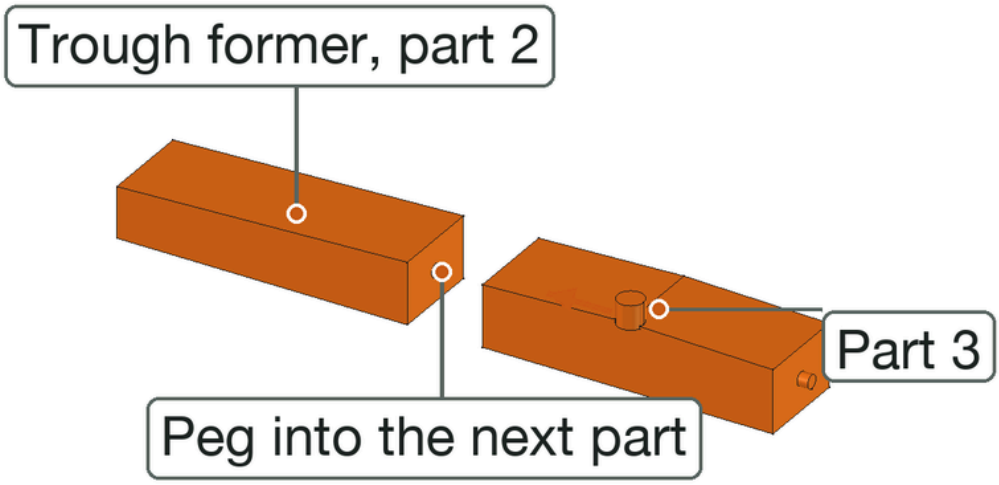
- **Printed pieces** have a nut built in: it goes into a pocket on the board side and slides under the hole; an M4 bolt comes through the board from outside.



- **Thin pieces** (gasket-groove beads) have 3 mm pins that push into 1/8 in holes.



- **Long pieces** are printed in parts that peg together; each part bolts down on its own.



- **Fits** are 0.15 mm a side, and drawn for 19.05 mm board (Home Depot's 3/4 in melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 3/4 in	about 0.4 m2	all the molds come out of 1 sheet, 4 x 8 ft
1x2 lumber (furring strip is fine)	1 x 8 ft 1x2 (1.4 m)	the cleats: they lie on the floor outside the walls
PETG	0.11 kg	sliced; everything printed for the machine fits in two 1 kg rolls
Paste wax and PVA	1 each	release
3/8 in quarter-round PVC molding	a few meters	the rounded outside edges

Hardware	Qty
M4 nut	21
M4 washer	21
M4 x 30 bolt	11
M4 x 45 bolt	10
M5 x 20 screw, greased (holds an insert)	2
M5 x 45 screw, greased (holds the face plate)	8
M6 x 45 screw, greased (holds the face plate)	6
1/2 in plain steel rod, greased (forms a beam bolt hole), 204 mm	8

Add about 10 % spare small bolts, nuts and washers. **Tools:** drill and brad-point bits (sizes in section 5), a square, clamps, a 7 mm nut driver, a 3 mm hex key, an awl.

3. Print

13 prints, 0.11 kg, about 8 hours (sliced; settings in the print guide). Print one corner bracket and one saddle first and check them on a board offcut. Press or slide the nuts in before the pieces go on the boards.

File	Qty	g	Time each
`beam/kit-corner-bracket.stl`	4	33	40m 3s
`beam/filler-back-edge-part1of3.stl`	1	17	1h 5m 47s
`beam/filler-back-edge-part2of3.stl`	1	17	1h 6m 9s
`beam/filler-back-edge-part3of3.stl`	1	17	1h 5m 49s
`beam/kit-dowel-stop.stl`	2	1	4m 19s
`tools/kit-pour-gauge.stl`	2	9	24m 33s
`tools/kit-screed-shoe.stl`	2	20	44m 42s

4. Cut the boards

Have the store rip the strips; crosscut at home. Write each board's name on its outside face, and mark the cast face (the melamine face the mix touches). The cleats are cut from 1x2 lumber, not the sheet; the bridges and struts come out of the sheet's offcuts.

Board	Qty	Size (mm)	Notes
Beam floor	1	630 x 264	
Beam top wall	1	550 x 165	
Beam bottom wall	1	550 x 165	
Beam end wall, left	1	146 x 165	
Beam end wall, right	1	146 x 165	
Beam cleat, top	1	550 x 38	1x2 lumber; flat on the floor
Beam cleat, bottom	1	550 x 38	1x2 lumber; flat on the floor
Beam cleat, left	1	146 x 38	1x2 lumber; flat on the floor
Beam cleat, right	1	146 x 38	1x2 lumber; flat on the floor

5. Mark and drill

1. Print the drilling templates (`templates/beam-templates.pdf`) at **actual size** and check the 100 mm square.
2. Tape each board's pages onto it: printed edges on the board's edges, grid lines lined up.
3. Awl through every crosshair. Pencil the dashed **pour line** on the walls.
4. Drill from the marked face, at the size the color legend gives. Pilot the big ones; the 63.5 mm holes are a 2-1/2 in hole saw.

The appendix has every board as a drawing and a table, if you would rather measure.

Hole	Drill	For
4.5 mm	11/64 in	back-edge, cleat, corner bracket
5.5 mm	7/32 in	M5 insert screw, M5 screw into the face plate
6.5 mm	17/64 in	M6 screw into the face plate
9.6 mm	25/64 in	3/8 dowel
13.1 mm	33/64 in	1/2 core rod

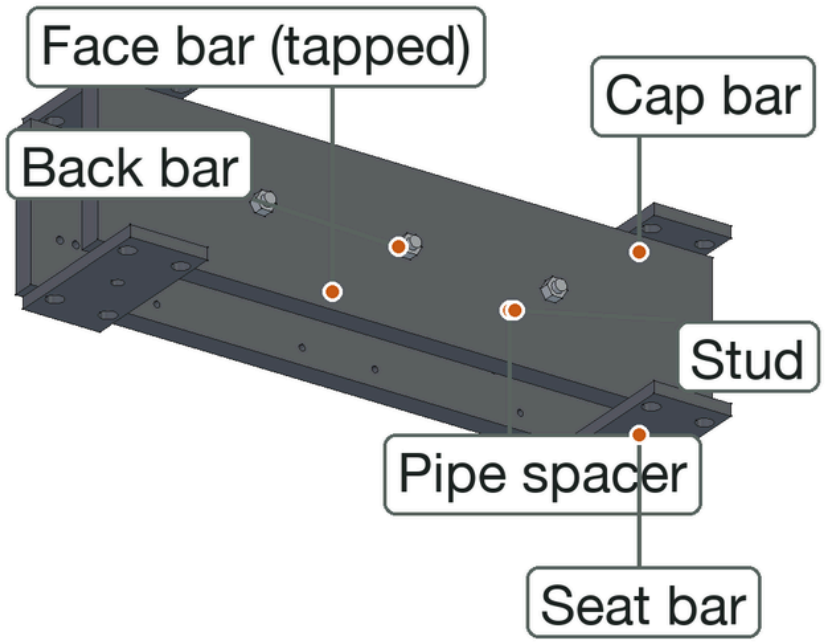
6. Wax

- 3-5 coats of paste wax on every face the mix touches, boards and removable printed pieces, then PVA. Don't wax pieces that stay in (core posts, cast-in parts): scuff and degrease them.
- Glue quarter-round into the floor-to-wall corners that make outside edges; silicone fillets elsewhere.
- Grease every screw, bolt and rod that comes out of the casting.

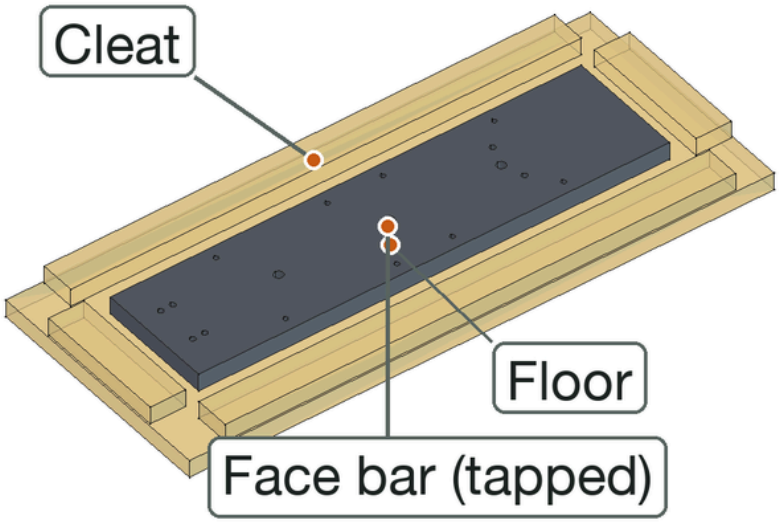
7. Assemble

Beam

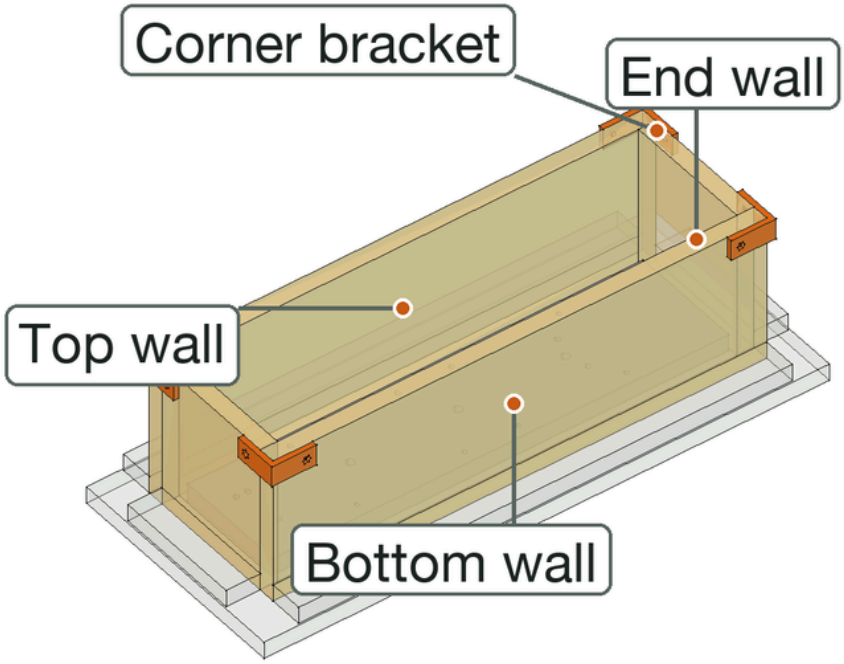
Cast front (rail) face down, back open. Bolt the steel together on the bench: the three studs into the face bar, a pipe spacer on each, the back bar over them and a nut on each.



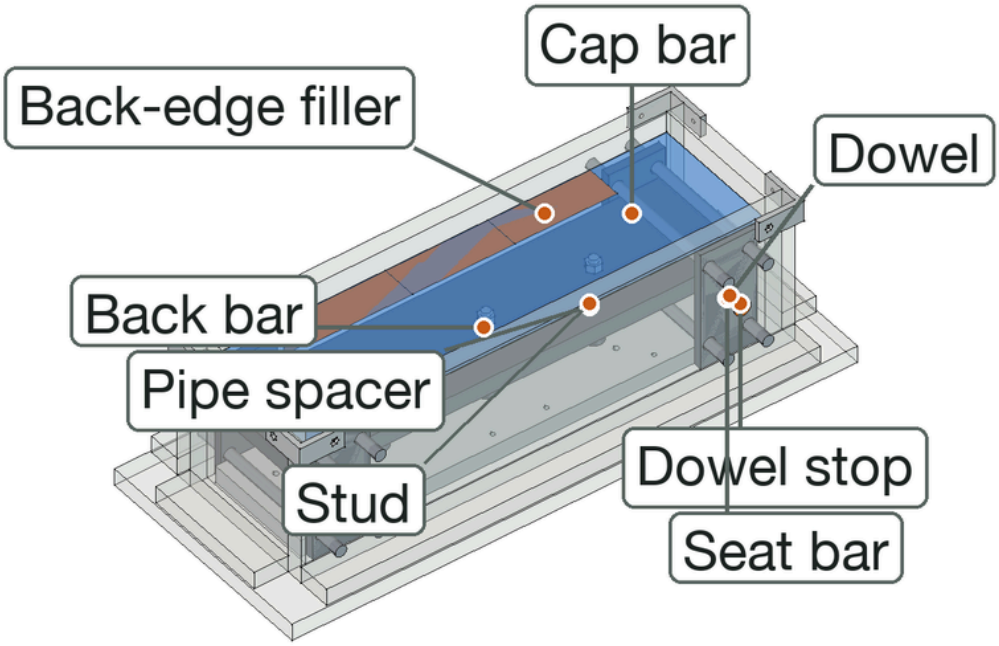
1. Cleats bolted down.



2. Walls against the cleats; four corner brackets.

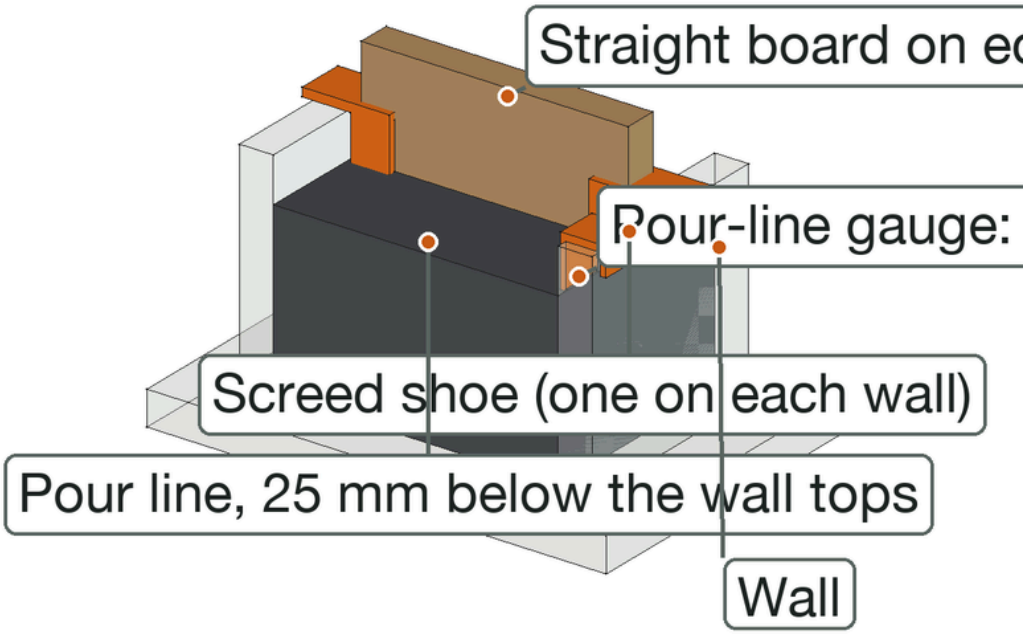


3. The steel in, face bar down: greased M5 and M6 screws up through the floor into its tapped holes; the back bar on its pipe spacers and studs; greased 1/2 in rods through the top wall, the cap bars, the seat bars and the bottom wall; back-edge filler on the top wall between the uprights; dowels on their stops. Fill to the line.



8. Pour

- Fill to the **pour line**, 25 mm below the wall tops (pencilled from the templates, or with the printed pour-line gauge clipped over a wall).
- Vibrate each layer until it stops bubbling; check the brackets halfway through.
- Strike the top off with a straight board in the two printed **screed shoes**: they hang it from the wall tops with its bottom edge on the pour line.



9. Take apart

1. After the full cure: bridges and saddles off, then corner brackets, then cleats; pull the walls off.
2. Screws, bolts and rods out from outside; lift the casting off the floor.
3. Unbolt and pull the formers and fillers.
4. Scrape and re-wax the boards; keep each mold's kit in a box. The kit is reused.

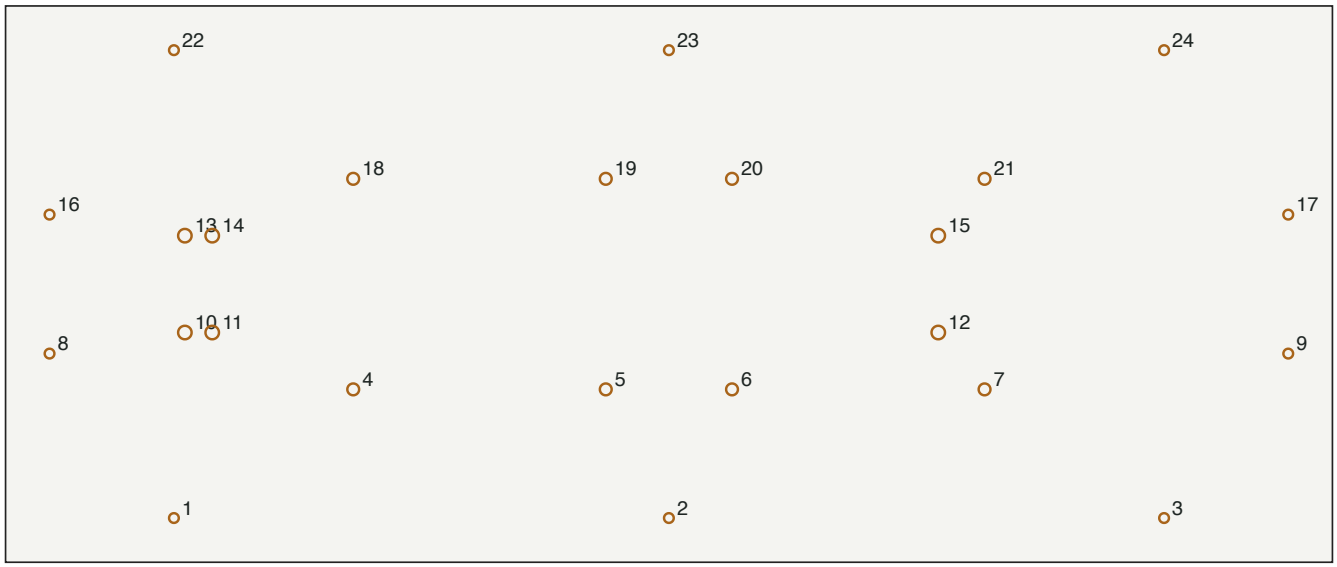
10. If something doesn't fit

Problem	Fix
Nut won't slide under the hole	Clean the pocket with a nut on a bolt; warm the nut
Bracket or saddle loose on the board	Board thinner than set: tape shim, or set `mold_board` and reprint
Pin won't go in	Run the 1/8 in bit through again; trim the pin tip
Foam core moves	Tighten its post bolt; carpet tape under it
Printed part won't go on or into a steel tube	File the weld seam; sand the part. Several tight: caliper the tube, put its sizes in `frame_tube` / `post_tube`, re-run and reprint
Printed part loose on a tube	A wrap of tape round the part
Leak at a corner	Silicone fillet inside before waxing; check the brackets and diagonals
Former stuck after curing	Bolts out first, then a dowel and a light tap

Appendix: hole maps

Every board as seen from the face you mark: x from the left edge, y from the bottom edge (for a wall, the edge on the floor). The templates carry the same holes at full size.

Beam floor

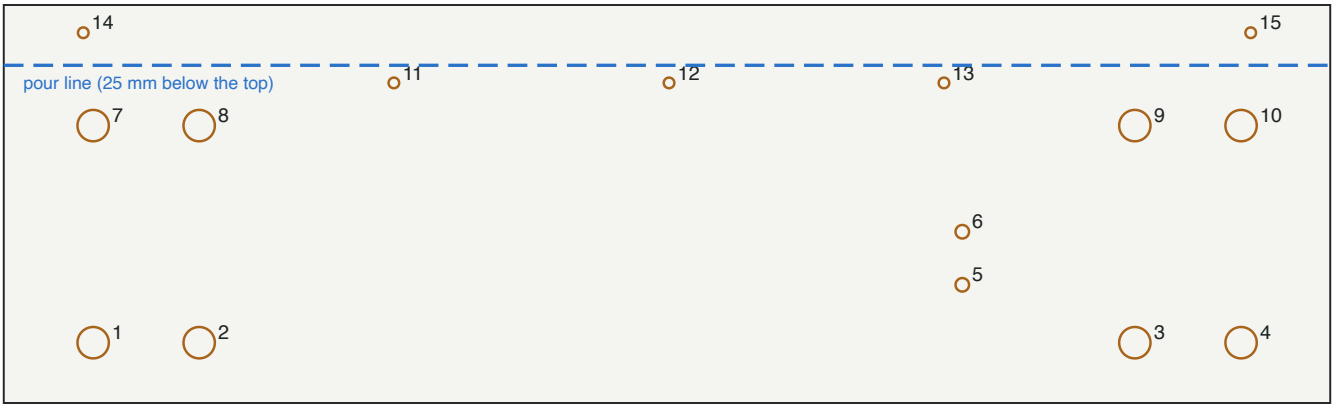


0,0 · 630 x 264 mm · seen from the cast side

#	x	y	Hole	For
1	80	21	4.5	cleat
2	315	21	4.5	cleat
3	550	21	4.5	cleat
4	165	82	5.5	M5 screw into the face plate
5	285	82	5.5	M5 screw into the face plate
6	345	82	5.5	M5 screw into the face plate
7	465	82	5.5	M5 screw into the face plate
8	21	99	4.5	cleat
9	610	99	4.5	cleat
10	85	109	6.5	M6 screw into the face plate
11	98	109	6.5	M6 screw into the face plate
12	443	109	6.5	M6 screw into the face plate
13	85	155	6.5	M6 screw into the face plate
14	98	155	6.5	M6 screw into the face plate
15	443	155	6.5	M6 screw into the face plate
16	21	165	4.5	cleat
17	610	165	4.5	cleat

#	x	y	Hole	For
18	165	182	5.5	M5 screw into the face plate
19	285	182	5.5	M5 screw into the face plate
20	345	182	5.5	M5 screw into the face plate
21	465	182	5.5	M5 screw into the face plate
22	80	243	4.5	cleat
23	315	243	4.5	cleat
24	550	243	4.5	cleat

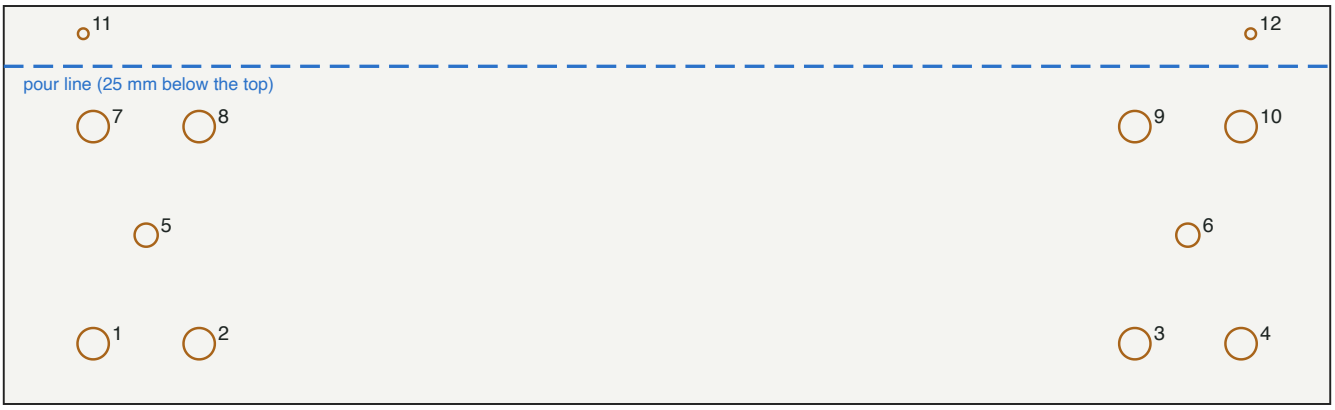
Beam top wall



0,0 · 550 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	469	25	13.1	1/2 core rod
4	513	25	13.1	1/2 core rod
5	398	49	5.5	M5 insert screw
6	398	71	5.5	M5 insert screw
7	37	115	13.1	1/2 core rod
8	81	115	13.1	1/2 core rod
9	469	115	13.1	1/2 core rod
10	513	115	13.1	1/2 core rod
11	162	133	4.5	back-edge
12	276	133	4.5	back-edge
13	390	133	4.5	back-edge
14	33	154	4.5	corner bracket
15	518	154	4.5	corner bracket

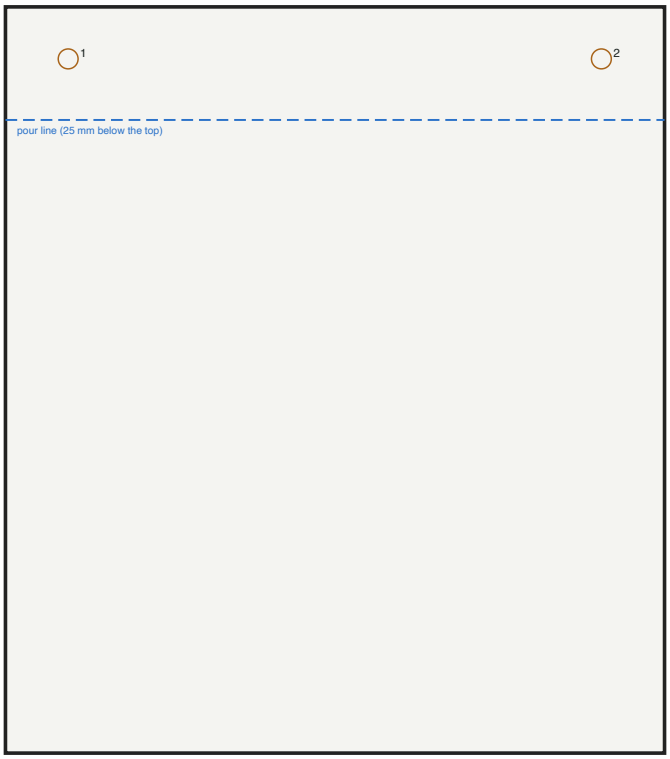
Beam bottom wall



0,0 · 550 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	469	25	13.1	1/2 core rod
4	513	25	13.1	1/2 core rod
5	59	70	9.6	3/8 dowel
6	491	70	9.6	3/8 dowel
7	37	115	13.1	1/2 core rod
8	81	115	13.1	1/2 core rod
9	469	115	13.1	1/2 core rod
10	513	115	13.1	1/2 core rod
11	33	154	4.5	corner bracket
12	518	154	4.5	corner bracket

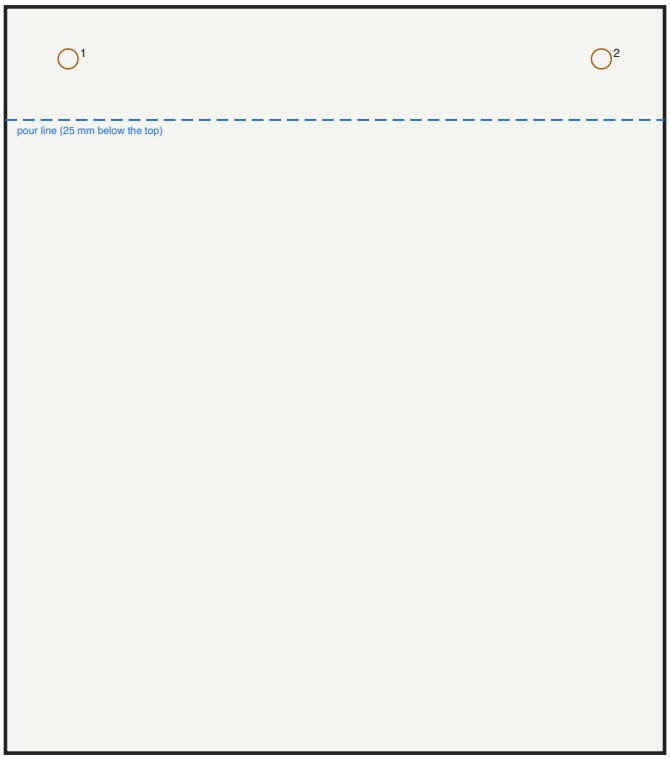
Beam end wall, left



0,0 · 146 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	14	154	4.5	corner bracket
2	132	154	4.5	corner bracket

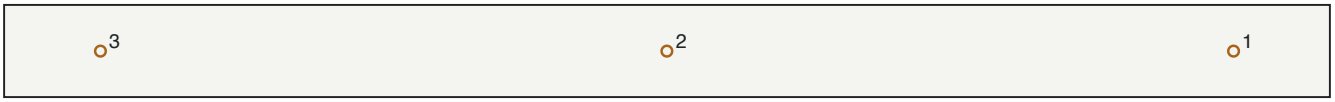
Beam end wall, right



0,0 · 146 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	14	154	4.5	corner bracket
2	132	154	4.5	corner bracket

Beam cleat, top



0,0 · 550 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	510	19	4.5	cleat
2	275	19	4.5	cleat
3	40	19	4.5	cleat

Beam cleat, bottom



0,0 · 550 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	510	19	4.5	cleat
2	275	19	4.5	cleat
3	40	19	4.5	cleat

Beam cleat, left



0,0 · 146 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	106	19	4.5	cleat
2	40	19	4.5	cleat

Beam cleat, right



0,0 · 146 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	106	19	4.5	cleat
2	40	19	4.5	cleat